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SMART CNG RESERVATION SYSTEM WITH SECURE SLOT BOOKING, DIGITAL PAYMENTS AND INTELLIGENT FUEL STATION MANAGEMENT

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ABSTRACT

Long lines at gas stations and severe traffic congestion are the results of the growing use of compressed natural gas (CNG) automobiles. Conventional manual booking techniques are cumbersome, ineffective, and opaque when it comes to handling reservations and station availability. An AI-Powered Smart CNG Reservation System with Secure Slot Booking, Digital Payments, and Intelligent Fuel Station Management is presented in this study as a solution to these problems.

The proposed system is a web-based platform developed using Java, JSP, Servlets, MySQL, JWT Authentication, and Google Gemini AI. The system enables users to reserve fueling slots in advance, perform secure online transactions, locate nearby CNG stations through map integration, and interact with an AI-powered chatbot for assistance. The platform incorporates three major modules: User, Pump Owner, and Administrator. Users can register, book slots, make payments, generate invoices, and track booking history. Pump owners can manage station availability, reservations, and customer interactions. Administrators monitor platform activities, manage registrations, and maintain operational transparency.

The integration of JWT-based authentication enhances security, while the Gemini AI chatbot improves user engagement through intelligent conversational support. Google Maps integration assists users in locating nearby CNG stations efficiently. The proposed solution reduces waiting time, improves fuel station utilization, enhances customer satisfaction, and streamlines fuel station management.

Experimental evaluation demonstrates improved reservation efficiency, secure access control, and effective queue management. Future enhancements include machine learning-based demand forecasting, mobile application support, and predictive slot allocation for optimized fuel station operations.

Keywords: CNG Reservation System, Smart Fuel Management, Slot Booking, JWT Authentication, Google Gemini AI, Digital Payment, Fuel Station Management, Web Application.

I. INTRODUCTION

One of the most affordable and eco-friendly alternative fuels for transportation is compressed natural gas, or CNG. The number of CNG-powered vehicles has grown dramatically in recent years due to rising fuel costs and growing environmental concerns. Due to the increased demand at CNG gas stations brought on by this quick expansion, there are now long lines, traffic jams, fuel waste, and disgruntled patrons.

Most existing CNG stations still rely on traditional manual methods for managing vehicle queues and fuel reservations. Customers are required to physically visit stations and wait for extended periods before refueling. Such approaches often result in inefficient utilization of station resources, poor customer experience, and lack of transparency in reservation management. Furthermore, users have limited information regarding station availability, waiting times, and nearby fueling locations.

An intelligent platform that can improve fuel station management and automate reservation processes is becoming more and more necessary as web technologies, artificial intelligence (AI), digital payment systems, and location-based services evolve. By allowing customers to reserve fuelling slots in advance, smart reservation systems can drastically cut down on wait times, increasing both consumer convenience and operational effectiveness.

This study suggests an AI-Powered Smart CNG Reservation System with Secure Slot Booking, Digital Payments, and Intelligent Fuel Station Management as a solution to these problems. The web-based platform is built with Java, JSP, Servlets, MySQL, and JWT Authentication. It offers an integrated environment where users can find local stations, book CNG filling times, make digital payments, create invoices, and get real-time help from an AI-powered chatbot.

The proposed system consists of three primary modules: User Module, Pump Owner Module, and Administrator Module. The User Module enables customers to register, login, search stations, reserve slots, make payments, and access booking history. The Pump Owner Module allows station operators to manage reservations, monitor customer activity, and maintain slot availability. The Administrator Module provides centralized control for monitoring users, approving pump registrations, and generating system reports.

To enhance user experience and platform security, the system integrates Google Maps services for location tracking, Google Gemini AI for conversational assistance, and JWT-based authentication for secure access control. These technologies collectively improve system usability, reliability, and scalability while ensuring secure communication between users and the platform.

The primary objectives of the proposed system are:

- To provide an efficient online platform for advance CNG slot booking.
- To reduce waiting time and congestion at CNG fueling stations.
- To employ JWT technology to offer secure user authentication and access control.
- To integrate digital payment facilities and automated invoice generation.

- To provide intelligent user support through AI-powered chatbot services.
- To assist users in locating nearby CNG stations using location-based services.
- To improve operational efficiency and transparency in fuel station management.

The proposed Smart CNG Reservation System aims to modernize traditional fueling operations by combining Artificial Intelligence, secure web technologies, location services, and reservation management within a unified platform. The system contributes to the development of smart transportation infrastructure by providing a scalable, efficient, and user-friendly solution for CNG station management and customer service.

II. METHODOLOGY

2. METHODOLOGY AND SYSTEM DESIGN

2.1 Proposed Methodology

The proposed AI-Powered Smart CNG Reservation System with Secure Slot Booking, Digital Payments and Intelligent Fuel Station Management is developed using Agile Software Development Life Cycle (SDLC) methodology. Agile methodology allows for iterative development, constant testing, regular feedback and easy implementation of new features. This ensures flexible, scalable, maintainable and improved software quality throughout the development process.

The system aims to automate CNG station booking operations with secure authentication, online slot booking, digital payment, location based services and AI empowered user assistance. The proposed solution is made up of multiple interconnected modules that work in tandem to provide a seamless reservation experience for users and efficient management tools for fuel station operators.

Technologies Utilized

Frontend Technologies

- HTML5
- CSS3
- JavaScript
- JSP (Java Server Pages)
- Bootstrap Framework

Backend Technologies

- Java
- Servlets
- JDBC

JWT Authentication

- MySQL Database Server

Artificial Intelligence Technologies

- Google Gemini AI
- Conversational Chatbot Integration

Location Services

- Google Maps API
- GPS-Based Station Location Services

Payment Technologies

- Online Payment Gateway Integration
- QR Code Based Payment Support

Development Tools

- Eclipse IDE
- Apache Tomcat Server
- MySQL Workbench
- GitHub
- Docker Containerization

2.2 System Design

The proposed Smart CNG Reservation System follows a three-tier architecture consisting of Presentation Layer, Business Logic Layer, and Data Layer.

Presentation Layer

The presentation layer provides a user-friendly interface through web pages developed using JSP, HTML, CSS, and JavaScript. This layer allows users, pump owners, and administrators to interact with the system through different dashboards and management portals.

Business Logic Layer

The business logic layer is implemented using Java Servlets and handles all core functionalities such as user authentication, slot booking, reservation management, payment processing, chatbot communication, and administrative operations.

Data Layer

The data layer utilizes MySQL Database for storing user records, booking details, station information, payment transactions, and administrative data. Database operations are performed using JDBC connectivity.

2.3 System Architecture

The architecture of the proposed Smart CNG Reservation System consists of several integrated modules that communicate with each other to perform reservation and management operations efficiently.

The process begins when a user accesses the web application through a browser. After successful registration and login, authenticated users can search nearby CNG stations using Google Maps integration. The user selects a preferred station and books an available fueling slot according to convenience.

Once the booking is confirmed, the payment module processes online transactions and generates an invoice for the user. All reservation details are stored securely in the database.

The Gemini AI-powered chatbot assists users by answering queries related to reservations, station availability, booking procedures, and general system information.

Pump owners can manage slot availability, monitor bookings, and analyze customer reservations through a dedicated dashboard. Administrators supervise the entire platform, approve station registrations, manage users, and monitor system performance.

JWT Authentication ensures secure communication and access control by validating user sessions and protecting sensitive information from unauthorized access.

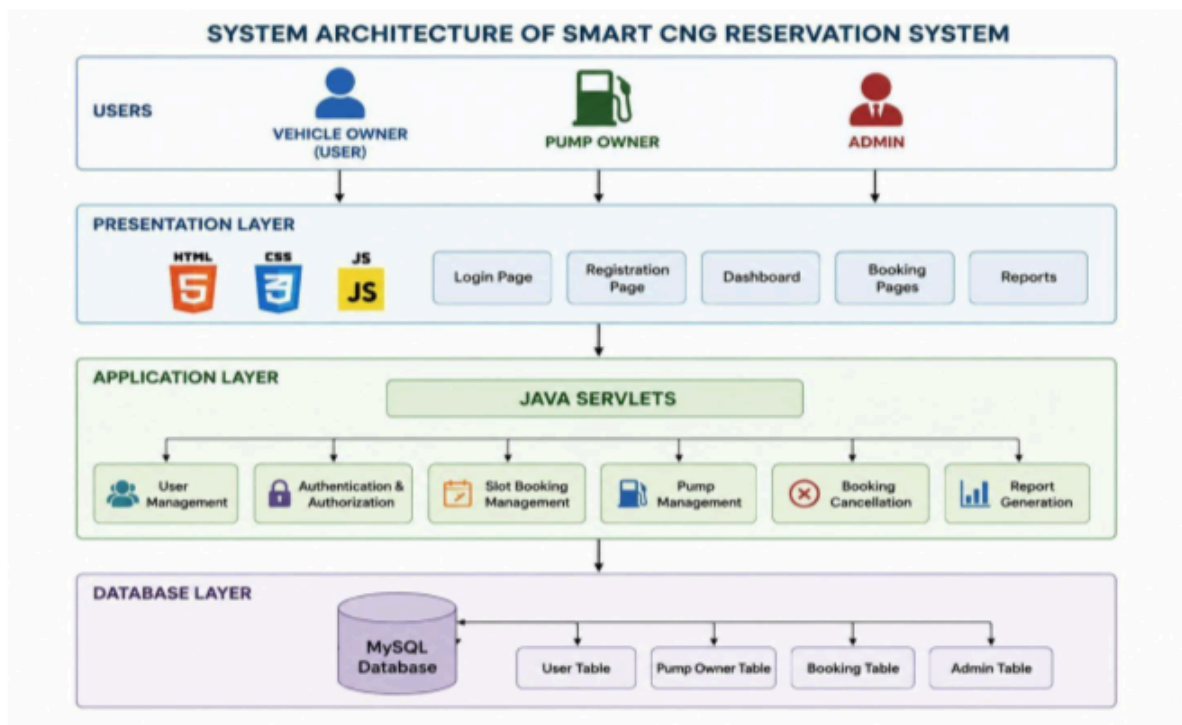


Figure 1. System Architecture

2.4 Functional Requirements

User Management

The system allows users to register, login, update profiles, and securely access system functionalities.

Slot Reservation

Users can search for available stations, view available slots, and reserve fueling appointments in advance.

Payment Processing

The platform supports online payment transactions and generates invoices after successful booking confirmation.

Location Tracking

Users can locate nearby CNG stations using integrated Google Maps services.

Chatbot Assistance

The AI-powered chatbot provides instant support and guidance regarding bookings, stations, and platform usage.

Pump Management

Pump owners can manage station details, monitor reservations, update slot availability, and view customer records.

Administrative Control

Administrators can manage users, approve pump registrations, monitor bookings, and generate reports.

Booking History

Users can access previous reservations, invoices, and transaction records through their dashboard.

2.5 Non-Functional Requirements

Performance

The system should process user requests and booking operations efficiently with minimal response time.

Security

JWT Authentication, session management, and secure database connectivity ensure protection against unauthorized access and data breaches.

Availability

The platform remains accessible to users whenever internet connectivity and server resources are available.

Scalability

The architecture supports future expansion including mobile applications, machine learning integration, and cloud deployment.

Reliability

The system maintains stable performance and accurate booking operations even under high user traffic conditions.

Usability

The web interface is designed to be user-friendly, intuitive, and accessible to users with minimal technical knowledge.

Maintainability

The modular architecture enables easy updates, bug fixes, and feature enhancements without affecting existing functionalities.

III. LITERATURE REVIEW

Several researchers have proposed reservation and booking systems for reducing waiting time and improving resource utilization.

The paper “Streamline Your Drive: Pre-booking and Pre-paid Parking Slots, Locating CNG Stations & EV Charging Stations” introduced a pre-booking mechanism that helps users reserve slots in advance and reduces congestion.

Another study, “Smart EV Charger Locator and Slot Booking Manager App for Chhattisgarh”, focused on real-time station location tracking and slot management for electric vehicle charging stations.

The research “Priority-based Pre-booking System for Electric Vehicle Charging Stations Enabling Emergency Vehicle Accessibility” proposed a priority allocation model to improve accessibility for emergency vehicles.

Several smart parking and reservation systems have also demonstrated the effectiveness of online booking, digital payments, and real-time monitoring in improving operational efficiency and customer satisfaction.

Although these systems provide booking and reservation facilities, most of them focus on parking management or EV charging infrastructure. Very limited research has been carried out specifically for CNG fuel station management with integrated AI assistance, secure authentication, digital payments, and real-time location services.

Research Gap Analysis

The following limitations were identified in existing systems:

- Lack of dedicated solutions for CNG station reservation management.
- Limited integration of AI-based user assistance.
- Absence of secure JWT-based authentication mechanisms.
- Insufficient support for online payments and invoice generation.
- Lack of centralized administration and pump management facilities.
- Limited use of real-time location services for nearby station discovery.

To overcome these limitations, the proposed AI-Powered Smart CNG Reservation System integrates secure slot booking, digital payments, Google Maps services, JWT authentication, and Gemini AI chatbot support within a single platform.

IV. SYSTEM ARCHITECTURE & DESIGN

The proposed AI-Powered Smart CNG Reservation System follows a three-tier architecture consisting of User, Pump Owner, and Admin modules connected through a centralized web server and MySQL database. Users can register, login, search nearby CNG stations, book fueling slots, make online payments, and interact with the AI chatbot. Pump owners manage station details, slot availability, and customer bookings. The Admin module monitors users, approves pump registrations, and maintains system operations. JWT Authentication ensures secure access, while Google Maps integration helps users locate nearby stations. All booking, payment, and user information is stored in the centralized database for efficient management and monitoring.

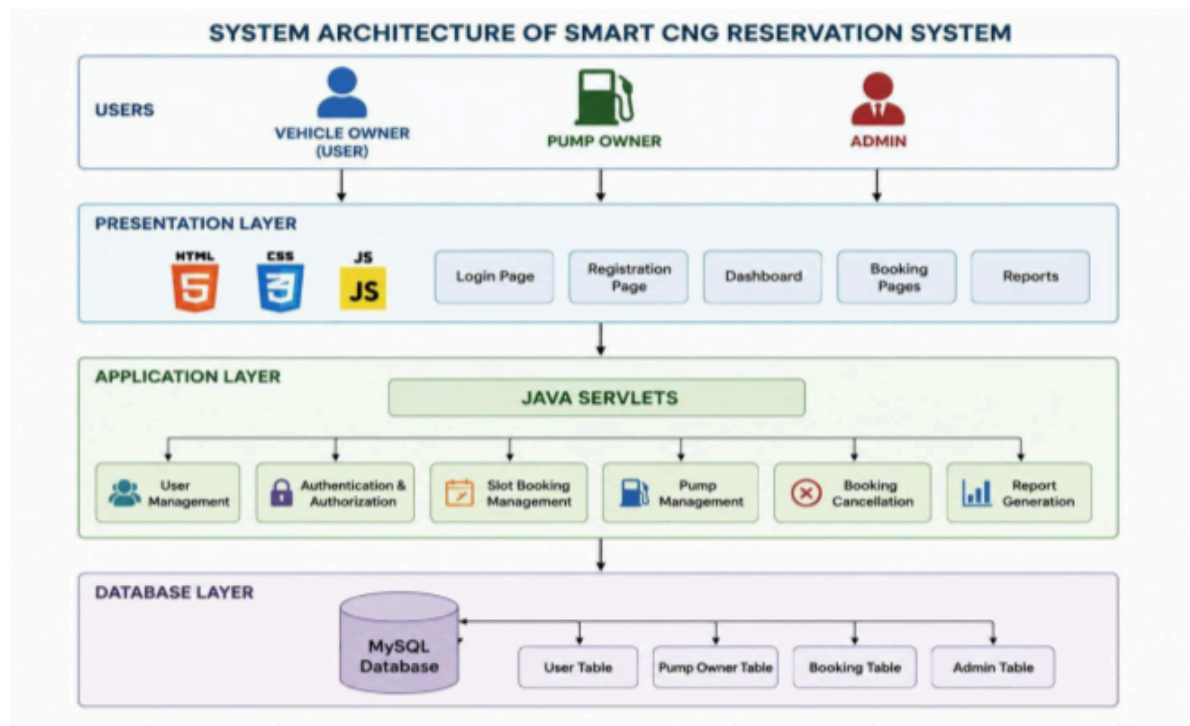


Figure 2: SYSTEM OVERVIEW DESIGN

V. RESULTS & DISCUSSION

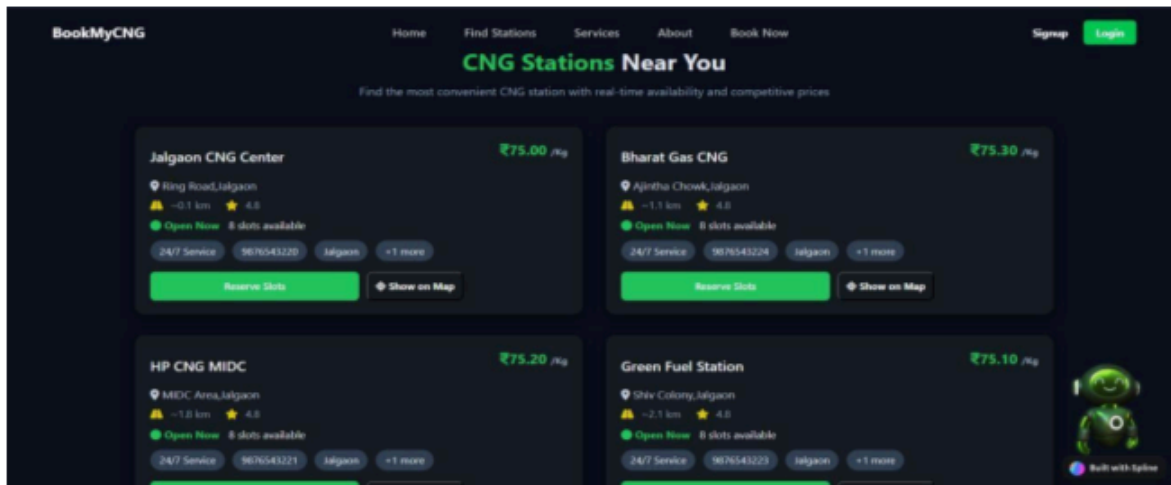
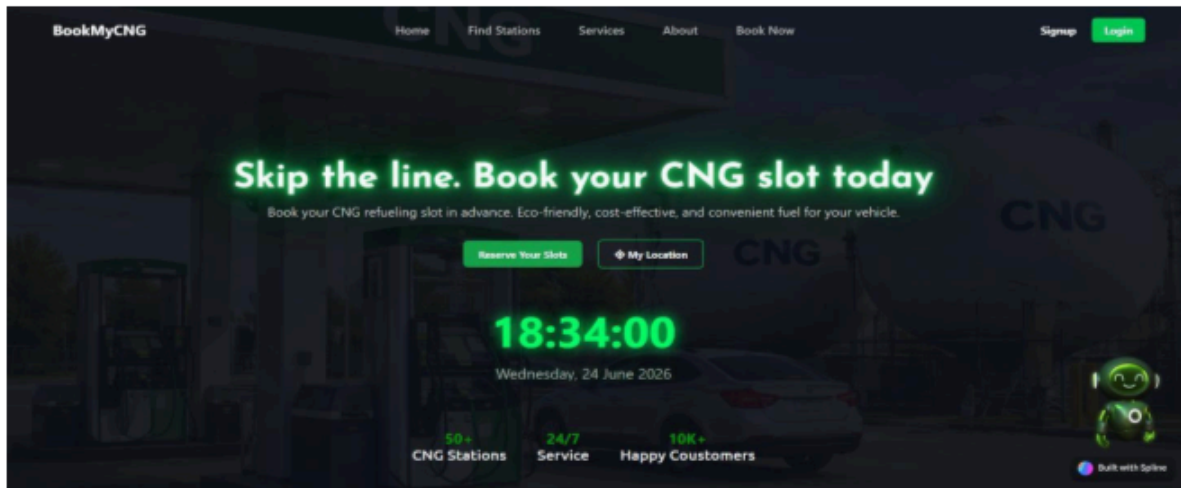
5.1 Results

The proposed AI-Powered Smart CNG Reservation System was successfully developed and tested using various booking and management scenarios. The system enables users to reserve CNG fueling slots online, make payments, locate nearby stations, and interact with the AI chatbot for assistance.

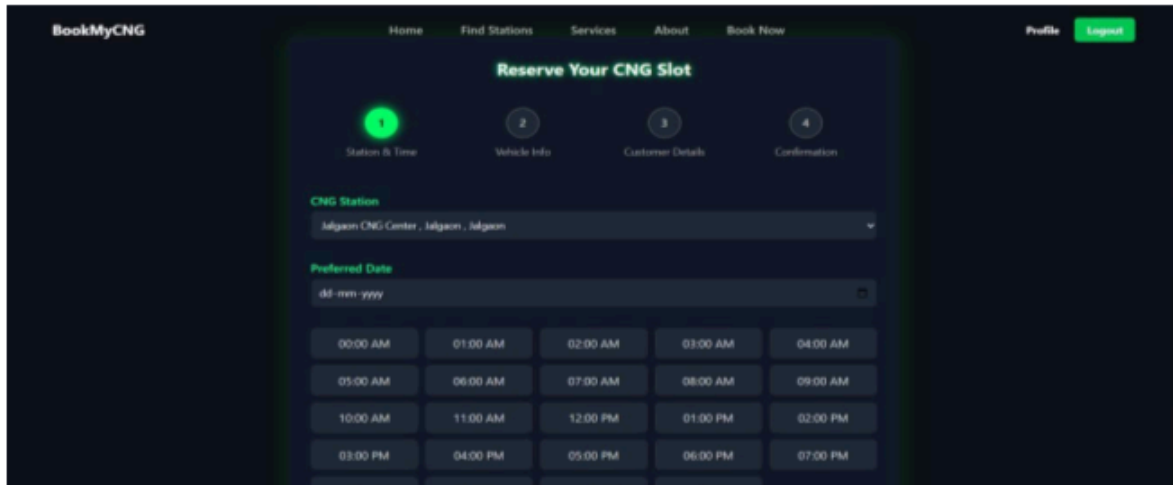
The User Module allows customers to register, login, book slots, view booking history, and generate invoices efficiently. The Pump Owner Module helps station operators manage reservations and slot availability. The Admin Module provides centralized monitoring of users, pumps, and bookings.

The integration of JWT Authentication ensures secure access control, while Google Maps and Gemini AI chatbot improve user experience and accessibility.

User Dashboard



Slot Booking Page



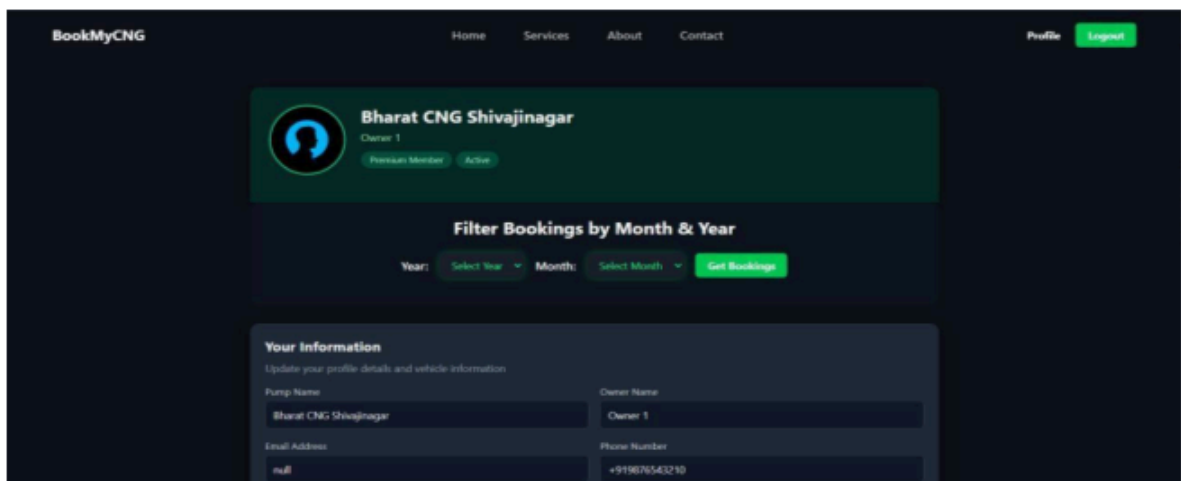
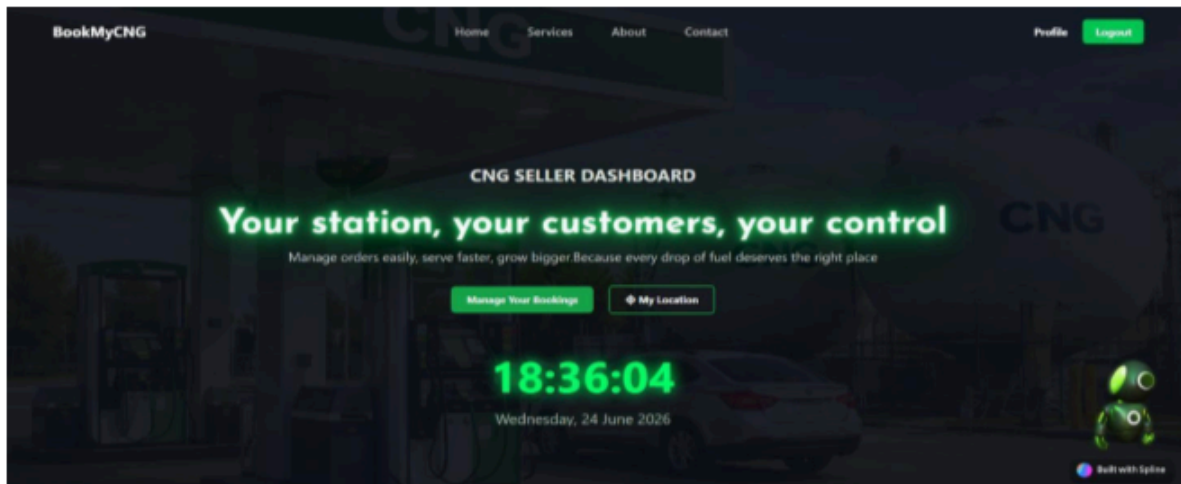
The screenshot shows the 'Reserve Your CNG Slot' page on the BookMyCNG app. The page has a dark blue background with white text. At the top, there's a navigation bar with 'Home', 'Find Stations', 'Services', 'About', and 'Book Now'. On the right, there's a 'Profile' link and a 'Logout' button. The main heading is 'Reserve Your CNG Slot'. Below it, there are four numbered steps: 1. Station & Time, 2. Vehicle Info, 3. Customer Details, and 4. Confirmation. Step 1 is currently active. Under 'CNG Station', there's a dropdown menu showing 'Jalgaon CNG Center, Jalgaon, Jalgaon'. Under 'Preferred Date', there's a date picker showing 'dd-mm-yyyy'. Below the date picker is a grid of time slots from 00:00 AM to 07:00 PM in 1-hour increments.

Payment Page

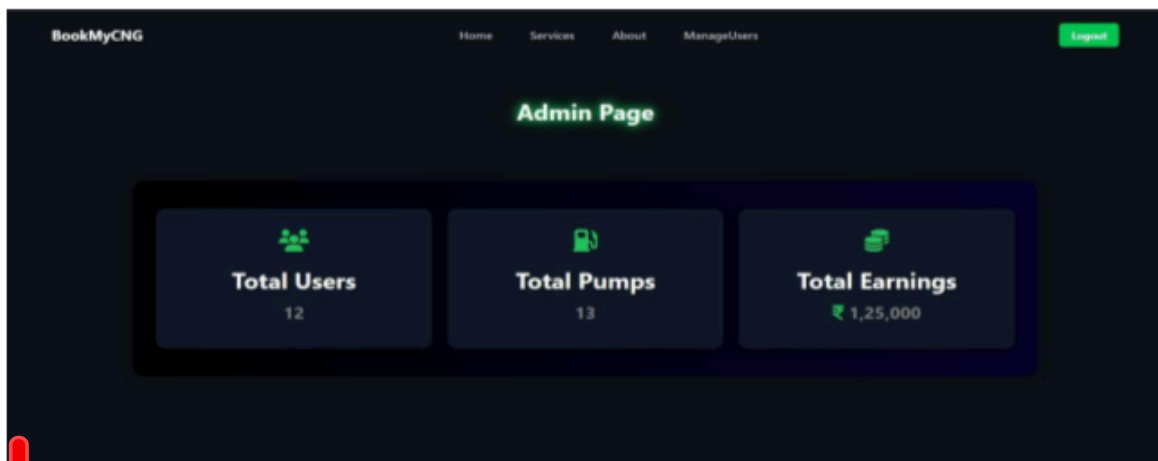


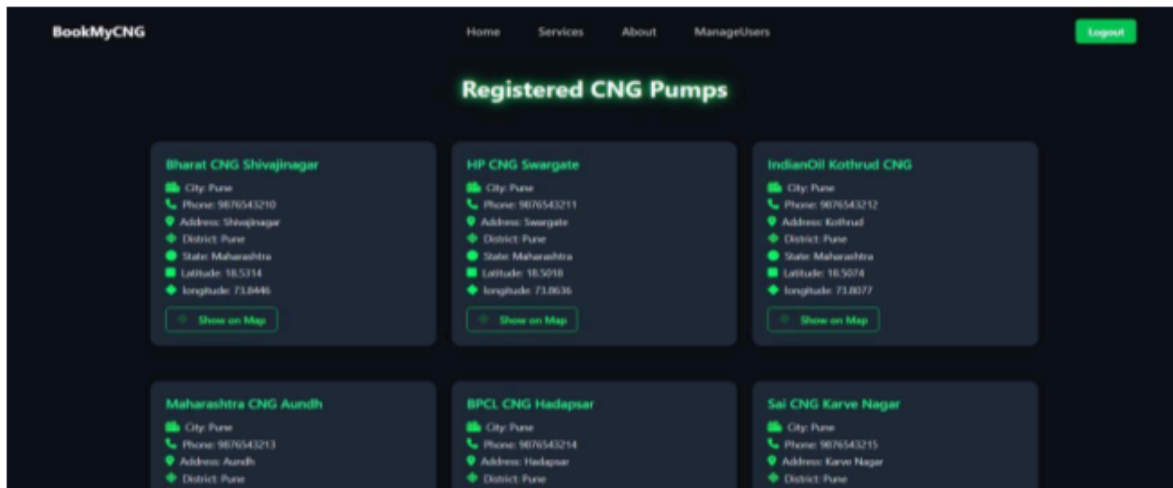
The screenshot shows the 'Confirmation' page on the BookMyCNG app. The page has a dark blue background with white text. At the top, there's a navigation bar with 'Home', 'Find Stations', 'Services', 'About', and 'Book Now'. On the right, there's a 'Profile' link and a 'Logout' button. The main heading is 'Confirmation'. Below it, there's a sub-heading 'Review your details and confirm your booking.' Under 'Your Order', there's a table showing the breakdown of costs: Platform Fees (₹ 30.00), Service charges (₹ 5.00), and Tax (₹ 6.30). The total is ₹ 41.30. There's a 'Download Receipt' button. Under 'Pay via UPI', there's a QR code and a prompt to 'Scan the QR code below to complete your payment'. At the bottom, there's a 'Previous' button and a 'Confirm Booking' button.

Pump Owner Dashboard



Admin Dashboard





5.2 Discussion

The experimental results show that the proposed system effectively reduces waiting time at CNG stations by enabling advance slot booking. Users can easily identify nearby stations, reserve convenient time slots, and complete payments online without standing in long queues.

The AI chatbot assists users by providing quick responses to common queries, while the dashboard modules help administrators and pump owners manage operations efficiently. The system improves transparency, resource utilization, and customer satisfaction through a centralized reservation platform.

Overall, the Smart CNG Reservation System provides a secure, reliable, and user-friendly solution for modern fuel station management.

VI. CONCLUSION AND FUTURE WORK

6.1 Conclusion

The proposed AI-Powered Smart CNG Reservation System provides an efficient and secure solution for managing CNG fuel station reservations. The system allows users to book fueling slots in advance, perform online payments, locate nearby stations, and receive assistance through an AI-powered chatbot. The integration of JWT Authentication enhances security, while Google Maps improves accessibility and user convenience.

The implementation of separate User, Pump Owner, and Admin modules ensures smooth management of reservations and station operations. The system successfully reduces waiting time, improves resource utilization, increases transparency, and enhances customer satisfaction. Overall, the proposed platform modernizes traditional CNG station management and provides a reliable solution for smart fuel reservation services.

6.2 Future Work

The system can be further enhanced by incorporating advanced technologies and additional functionalities. Future improvements may include:

- Development of a mobile application for Android and iOS platforms.
- Machine Learning based demand prediction and smart slot allocation.
- Real-time queue monitoring and waiting time estimation.
- Integration of live fuel availability information.
- Online payment gateway enhancements with multiple payment options.
- Cloud-based deployment for improved scalability and performance.
- Advanced analytics and reporting dashboards for station owners.
- Multilingual AI chatbot support for better user interaction.

These enhancements will further improve operational efficiency and provide a more intelligent and user-friendly fuel station management platform.

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